NTIRE 2026 Mobile Real-World Image Super-Resolution Challenge Factsheet -06_IMAG2006-

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1. Factsheet Information

The factsheet should contain the following information. Most importantly, you should describe your method in detail. The training strategy (optimization method, learning rate schedule, and other parameters such as batch size, and patch size) and training data (information about the additional training data) should also be explained in detail.

1.1. Team details

- Team name: IMAG2006
- Team leader name: Xinzhe Zhu
- Team leader address, phone number, and email: Nanjing University of Science and Technology, 18192332465, xzzhu@njust.edu.cn
- Rest of the team members: Fengkai Zhang, Junyang Chen, Congyu Wang, Yixin Yang, Zhaorun Zhou, Jiangxin Dong, Jinshan Pan
- Team website URL (if any): NJUST_IMAG
- Affiliation: Nanjing University of Science and Technology
- Affiliation of the team and/or team members with NTIRE 2026 sponsors (check the workshop website): None
- User names and entries on the NTIRE 2026 Codabench competitions (development/validation and testing phases): xzzhu
- Best scoring entries of the team during the testing phase: CLIPIQA 0.8777 MUSIQ 73.7207 ID 634557
- Link to the codes/executables of the solution(s): NTIRE2026_IMAG2006

1.2. Method details

Description. The proposed solution is shown in Fig. 1. Our method is mainly built upon OMGSR [5], a one-step super-resolution approach based on diffusion models. It achieves low-quality (LQ) image injection for restoration by only adjusting the diffusion timestep corresponding to the LQ input.

During training, we replace the original DINOv3-based DISTS loss in OMGSR with the LPIPS loss [8], denoted as $\mathcal{L}_{\text{LPIPS}}$, and the DISTS loss [2], denoted as $\mathcal{L}_{\text{DISTS}}$, to provide richer perceptual supervision and improve the perceptual relevance of the training objective.

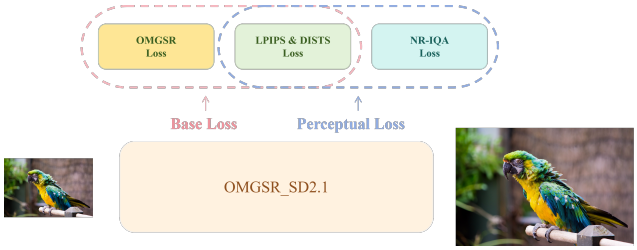

Figure 1. Overview of the proposed method.

$$\mathcal{L}_{\text{base}} = \lambda_1 \mathcal{L}_{\text{LRR}} + \lambda_2 \mathcal{L}_{\text{L1}} + \lambda_3 \mathcal{L}_{\text{GAN}} + \lambda_4 \mathcal{L}_{\text{LPIPS}} + \lambda_5 \mathcal{L}_{\text{DISTS}}, \tag{1}$$

where $\mathcal{L}_{\text{LRR}}$ is used to align the LQ input at the specified timestep with the noised HQ target, $\mathcal{L}_{\text{L1}}$ is introduced to preserve pixel-wise fidelity, and $\mathcal{L}_{\text{GAN}}$ is employed to enhance visual realism.

Since OMGSR is a one-step diffusion method based on a pre-trained model, it can preserve the rich prior knowledge of the pre-trained backbone. To guide the model toward producing images with stronger visual perception, We follow [7] and incorporate no-reference image quality assessment (NR-IQA) losses [3, 4, 6] as additional supervision during training.

The corresponding training objective is given by:

$$\mathcal{L}_{\text{NR-IQA}} = \lambda_6 \mathcal{L}_{\text{MUSIQ}} + \lambda_7 \mathcal{L}_{\text{CLIPIQA}} + \lambda_8 \mathcal{L}_{\text{MANIQA}}, \tag{2}$$

where $\mathcal{L}_{\text{MUSIQ}}$, $\mathcal{L}_{\text{CLIPIQA}}$, and $\mathcal{L}_{\text{MANIQA}}$ are all differentiable and therefore can be used as perceptual supervision signals. For metrics where higher scores indicate better perceptual quality, we first normalize the scores to $[0, 1]$ and then transform them into minimization objectives by using

$1 - s$, where s is the normalized score. Their implementations are mainly based on the `pyiqa` library.

The total training loss is formulated as:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{base}} + \mathcal{L}_{\text{NR-IQA}} \tag{3}$$

Implementation Details. We train the model using the AdamW optimizer with $\beta_1 = 0.9$, $\beta_2 = 0.999$, and a learning rate of 5×10^{-5}. The ground-truth images are cropped into 512×512 patches. Training is conducted for 6,000 iterations with a batch size of 4. The loss weights are set as $\lambda_1 = 5$, $\lambda_2 = 0.5$, $\lambda_3 = 0.5$, $\lambda_4 = 3$, $\lambda_5 = 3$, $\lambda_6 = 0.05$, $\lambda_7 = 2$, and $\lambda_8 = 1$. The VAE rank is set to 16, while the UNet rank is set to 32. Following FaithDiff [1], we use the same training datasets, including DIV2K, LSDIR, and other high-quality datasets.

2. Other details

- Planned submission of a solution(s) description paper at NTIRE 2026 workshop.
  It depends on the smooth progress of the follow-up plan and the competition results.
- General comments and impressions of the NTIRE 2026 challenge.
  We believe that the evaluation of the perception track requires more judging mechanisms, which is indeed a problem currently faced by the field of image restoration.
- What do you expect from a new challenge in image restoration, enhancement and manipulation?
  We hope the competition evaluation dataset can be more diverse and that the system can be smarter to make it easier for participants to view the rankings.
- Other comments: encountered difficulties, fairness of the challenge, proposed subcategories, proposed evaluation method(s), etc.

References

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